

Bayesian Uncertainty Estimation in Landmark Localization using Convolutional Gaussian Processes

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Abstract. Landmark localization is an important step in image analysis, where the clinical definition of a landmark can be ambiguous, leading to a practical necessity for model uncertainty quantification that is rigorous and trustworthy. In this paper, we present the first Bayesian framework using Gaussian processes to capture both dataset-level landmark ambiguity and sample-level model uncertainty. Our proposed two-stage approach includes a deep learning based U-Net for coarse predictions, followed by a convolutional Gaussian process (CGP) for fine-grained predictions with uncertainty estimates, learning covariance matrices rather than using a pre-defined covariance matrix. Our Bayesian approach yields a more rigorous quantification of uncertainty compared to deep learning-based uncertainty estimation techniques, whilst still achieving comparable localization accuracy. Our results suggest that CGPs can better model the inherent uncertainties in landmark localization tasks and provide more reliable confidence estimates, making it a promising direction for future research.

Keywords: Landmark Localization · Gaussian Process · Uncertainty · Bayesian

1 Introduction

Automatic landmark localization is a critical component of medical image analysis, encompassing tasks such as image segmentation [2] and image registration [9, 15]. However, defining anatomical landmarks with precision can be a challenging endeavor, especially for those situated on smooth edges, such as the corner of the jaw, in contrast to distinctive anatomical structures like the incisor tip [21]. In this regard, it would be preferable for a model to predict the distribution of likely landmark locations rather than a single point estimate. Quantifying the uncertainty associated with landmark predictions is crucial in a clinical setting, where explainability is essential and there is a need for human intervention to correct highly uncertain predictions [6]. Further, it is imperative that the uncertainty quantification can be trusted to represent a mathematically rigorous and

transparent uncertainty of the model, which is rarely the case for deep learning based approaches.

Current state-of-the-art (SOTA) approaches for landmark localization are deep-learning based, the most popular paradigm being heatmap regression [25, 17, 8]. Heatmap regression models perform landmark localization by learning an image-to-image function, where the target is a heatmap of a Gaussian function centered on the landmark of interest. It is noteworthy that conventional training methodologies in heatmap regression models fix the variance of the Gaussian function, preventing the model from modifying its output to reflect elevated or diminished uncertainty in its predictions.

Thaler *et al.* incorporated aleatoric uncertainty directly during training by learning anisotropic Gaussian heatmaps for each landmark. The study demonstrated that the learned heatmap shapes correspond to inter-observer variability from multiple annotators [21]. However, this method only models the homoscedastic aleatoric uncertainty of the dataset, whereby a single covariance matrix is learned over the entire dataset for each landmark during training. At inference, a Gaussian function is fitted to each individual prediction to model heteroscedastic aleatoric uncertainty. Nonetheless, this measure heavily depends on the learned homoscedastic uncertainty, and its post-hoc nature makes it challenging to rely on for a true reflection of model uncertainty. Other existing approaches to estimate landmark uncertainty include using the variance between an ensemble of convolutional neural networks (CNNs)[20], and utilizing Bayesian CNNs to predict confidence intervals [12]. However, these methods rely on deep neural networks, which have been notoriously criticized for their unreliability in terms of calibration [5] and interpretability.

In this study, we depart from the conventional practice of utilizing deep learning and instead employ a Bayesian methodology for landmark localization, relying on Gaussian processes (GPs). GPs are nonparametric statistical models which are robust to both the presence of noisy data and overfitting, even in low-data regimes which can prove challenging for neural network-based techniques [18]. Specifically, we use convolutional Gaussian processes (CGPs), which offer an attractive alternative to deep neural networks for the task of landmark localization. CGPs are constructed using a covariance function which is heavily inspired by the efficient convolutional structure of the kernels used in CNNs [24]. CGPs offer us a mathematically rigorous Bayesian framework for predicting the distribution of likely landmark locations and quantifying model uncertainty. The intrinsically Gaussian nature of the uncertainty estimates generated by CGPs render them an intriguing alternative to the conventional deep learning approaches that aim to predict a Gaussian heatmap (either fixed or learned).

Our contributions are threefold: **(1)** A two stage coarse-to-fine approach using a multi-task CGP for fine predictions and uncertainty estimates, which is the first of its kind for regression. **(2)** A novel approach to initialize inducing patches for the CGP using first-stage prediction information. **(3)** Evaluation of CGP uncertainty estimates against a deep learning baseline CNN method.

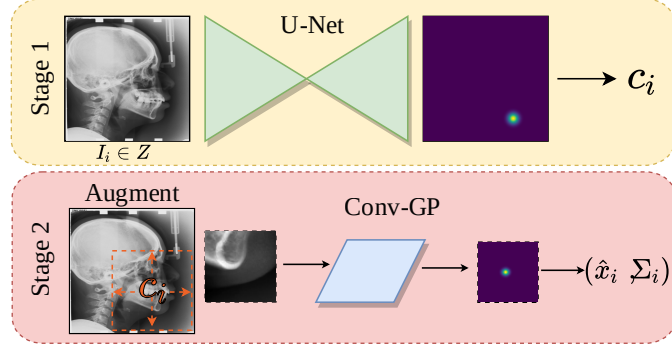


Fig. 1. Overview of our two stage coarse-to-fine framework. We utilize deep learning in Stage 1 to obtain coarse predictions, and refine them in Stage 2 with a convolutional Gaussian process and obtain an uncertainty estimate (covariance).

2 Methods

Due to limitations with the scalability of convolutional Gaussian processes (CGPs), we use a two stage coarse-to-fine approach, outlined in Figure 1. The first stage uses a CNN to obtain a coarse prediction of the landmark location. Then, the CGP predicts the final landmark distribution using the corresponding cropped patch of the image.

2.1 Stage 1: Coarse Prediction using U-Net

To obtain our coarse predictions, we design a vanilla U-Net following the template of the original U-Net [19] and nnU-Net [7]. The U-Net follows the standard configuration of two blocks per resolution layer, with each block consisting of a 3×3 convolution, instance normalization [22], and leaky ReLU (negative slope, 0.01). Downsampling is achieved through strided convolutions and upsampling through transposed convolutions. The initial number of feature maps is set to 32, doubling with each downsample to a maximum of 512 and halving at each upsample step. We automatically configure the number of resolution layers by adding encoder steps until any dimension of the feature map resolution hits a minimum of 4. The objective for the model is to learn a Gaussian heatmap image for each landmark, with the centre of the heatmap on the target landmark. For a landmark i with 2D coordinate position $\mathbf{x}_i$, the 2D heatmap image is defined as the 2D Gaussian function:

$$g_i(\mathbf{x} \parallel \boldsymbol{\mu} = \mathbf{x}_i; \sigma) = \frac{1}{(2\pi)\sigma^2} \exp\left(-\frac{\|\mathbf{x} - \boldsymbol{\mu}\|_2^2}{2\sigma^2}\right), \quad (1)$$

where $\mathbf{x}$ is the 2D coordinate vector of each pixel and σ is a user-defined standard deviation. The network learns weights $\mathbf{w}$ and biases $\mathbf{b}$ to predict the heatmap

$h_i(\mathbf{x}; \mathbf{w}, \mathbf{b})$. The objective function is the mean squared error (MSE) between the Gaussian target heatmap and the predicted heatmap. We train on images of size $H \times W$ at this stage, obtaining the coarse predictions of our landmarks, $\mathbf{C}$. For each landmark $\mathbf{x}_i$, we obtain $\mathbf{c}_i$ by selecting the pixel with the highest activation in each heatmap.

2.2 Stage 2: Fine Prediction using CGP

Following the initial prediction, we extract cropped patches of size $H' \times W'$, ($H' < H$, $W' < W$) around each image using our *Stage 1* predictions, $\mathbf{C}$. We then use a multi-task CGP for the final, sub-pixel prediction and uncertainty estimate.

Multi-task Convolutional Gaussian Processes Gaussian processes (GPs) are powerful nonparametric Bayesian models, gaining popularity due to their ability to provide a rigorous quantification of predictive uncertainty. A GP defines a distribution over functions and is completely specified by a covariance function $k(\cdot, \cdot)$ and a mean function which is commonly assumed to be zero. For an input $\mathbf{I} \in \mathbb{R}^{D_{\text{in}}}$ (e.g. an image), a GP is denoted by $u(\mathbf{I}) \sim \mathcal{GP}(0, k(\mathbf{I}, \mathbf{I}'))$ [18]. The choice of covariance function (also called the *kernel*) affects the variation of the function over the input domain, and many options for the kernel exist. The CNN-inspired *image convolutional kernel* [24] has become the standard tool for applying GPs to computer vision tasks. A GP with this kernel (i.e. a CGP) can be written as,

$$u \sim \mathcal{GP} \left(0, \sum_{p=1}^P \sum_{p'=1}^P k_g \left(\mathbf{I}^{[p]}, \mathbf{I}^{[p']} \right) \right), \quad (2)$$

where k_g is a base kernel (e.g. RBF or Matérn) which generates a real-valued response value for a square patch of pixels within the image $\mathbf{I}$. P denotes the total number of patches we can extract from our input image, and is therefore determined by the patch size, which we define to be (5×5) . Specifically, we use the weighted kernel proposed by the authors, whereby each patch is additionally assigned a learnable weighting parameter. As in the original work, we take a stochastic variational approach to performing inference, using a set of *inducing patches*. The intuition behind the overall approach is that the inducing patches can be considered as analogous to the filters in a conventional CNN. For further details on the approach we refer the reader to the original work [24] and material on variational inducing points-based inference in GPs [13].

Typically, CGPs are used with a single output. However, for our problem setting, we require a multi-output GP as we wish to predict a 2D coordinate associated with each landmark. To achieve this, we firstly model each output using independent GPs, each with their own separate convolutional kernel, sharing the inducing patches across both outputs. We instantaneously mix the outputs of these two GPs $\mathbf{u} \in \mathbb{R}^2$ using the *linear model of coregionalization* (LMC) [1,

10], such that $\mathbf{f} = \mathbf{W}\mathbf{u}$, where $\mathbf{f} \in \mathbb{R}^2$ are our correlated outputs, and $\mathbf{W} \in \mathbb{R}^{2 \times 2}$ is a learnable mixing matrix.

As this is a regression problem, we use a Gaussian likelihood with independent likelihood noise for each spatial dimension, denoted by l_x and l_y . Therefore, for each prediction we obtain the final sub-pixel coordinate prediction $\hat{\mathbf{x}}_i$ from the CGP mean prediction $\mathbf{m}_i$ and a covariance matrix Σ_i , which is the summation of the LMC covariance and the likelihood noise:

$$\hat{\mathbf{x}}_i = \mathbf{m}_i, \quad \Sigma_i = \mathbf{W}\mathbf{u} + \begin{pmatrix} l_x & 0 \\ 0 & l_y \end{pmatrix}. \quad (3)$$

Inducing Patch Initialization To initialize the inducing patches, we introduce a bias towards inducing patches proximal to the anticipated landmark location, given by *Stage 1*. We do so because regions in the image near the landmark tend to contain more salient information about its location as compared to regions further away. Therefore, this initialisation sets up a simpler optimisation problem as opposed to choosing the initial patches purely at random. For each inducing patch P , we randomly select an image from the training set and sample a 5×5 patch, selecting a patch centre point P_c by sampling from a Gaussian distribution over the image:

$$P_c \sim g(\mu = \mathbf{c}_i; \sigma), \quad (4)$$

where g is a 2D Gaussian Distribution (see Eq. (1)), $\mathbf{c}_i$ is the coarse prediction for landmark $\mathbf{x}_i$ from *Stage 1*, and σ is defined by the user.

Neural Network Baseline To serve as a deep learning baseline for *Stage 2* predictions, we replace the CGP with a compact U-Net architecture. Specifically, we adopt the identical design to the U-Net used in *Stage 1*, with the only difference being a smaller number of layers (due to the use of smaller 64×64 input images). To obtain a comparable likelihood distribution to the CGP, we follow the approach by Thaler *et al* [21] to fit a 2D Gaussian function to the heatmap prediction, robust least squares fitting method [3], obtaining the sub-pixel mean prediction $\hat{\mathbf{x}}_i$ and covariance matrix Σ_i .

Training Procedure We train using mini-batching, maximising the evidence lower bound (ELBO) at each iteration step (or the MSE for the CNN baseline). At each step, we extract a batch of $H' \times W'$ patches from the image batch, centred around the *Stage 1* predictions, $\mathbf{C}$. To prevent overfitting and improve robustness, we implement data augmentation by randomly selecting a new centre point for each patch, $\hat{\mathbf{c}}_i$, each time the data is seen using the following equation:

$$\hat{\mathbf{c}}_i = \mathbf{c}_i + \mathbf{u}, \quad \mathbf{u} \sim U(-D, D)^2, \quad (5)$$

where U is Uniform distribution and $D < \frac{H'}{2}, \frac{W'}{2}$, ensuring the *Stage 1* prediction $\mathbf{c}_i$ is always present in the image patch.

3 Experiments and Analysis

3.1 Dataset

We show results on the publicly available Cephalometric dataset [23], which consists of 400 cephalograms with 19 landmarks, each annotated by a senior and junior expert. We select 3 landmarks representing various difficulty levels: the tip of the chin, the corner of the jaw, and the tip of the incisor (exemplified in Figure 2). The images are a resolution of 1935×2400 pixels, where each pixel represents 0.1mm. The images are resized to a resolution of 512×512 pixels, and the final sub-pixel coordinate predictions are scaled to the original resolution. We report results of a 4-fold cross validation (CV) over all 400 images, using the junior annotations only, following convention [14, 21, 16].

3.2 Metrics

To evaluate the quality of uncertainty estimates in our predictive model, we report the negative log predictive density (NLPD):

$$NLPD = -\log p(\mathbf{y}_i | \mathbf{X}_i, \boldsymbol{\Sigma}_i) = -\sum_{j=1}^N \log p\left(y_i^{(j)} | \hat{\mathbf{x}}_i^{(j)}, \boldsymbol{\Sigma}_i^{(j)}\right), \quad (6)$$

where $\hat{\mathbf{x}}_i^{(j)}$ is the predicted coordinate for landmark i in image j , $\mathbf{y}_i$ represents the target landmarks for landmark i , $\mathbf{X}_i$ the predicted point-wise estimates, and $\boldsymbol{\Sigma}_i$ the predicted covariances. The NLPD quantifies the likelihood that the predictive distribution contains the true landmark location. Lower NLPD values indicate a better model fit to the data, and thus better uncertainty estimates.

For completeness, we evaluate localization performance using point-to-point error, defined by the Euclidean distance from the predicted coordinate for landmark i in image j , $\hat{\mathbf{x}}_i^{(j)}$, to the target coordinate $\mathbf{x}_i^{(j)}$: $PE(\mathbf{x}_i^{(j)}) = \|\mathbf{x}_i^{(j)} - \hat{\mathbf{x}}_i^{(j)}\|$.

3.3 Implementation Details

All hyperparameter tuning was performed on the first fold of landmark L_1 , the tip of the incisor.

Stage 1: For the target Gaussian heatmap in Eq. (1), we use $\sigma = 8$. We train for 500 epochs using stochastic gradient descent with an initial learning rate of 0.01, decaying it using the ‘poly’ scheme, $(1 - epoch/epoch_{max})^{0.9}$ [4]. One epoch consists of 150 mini-batches, where each mini-batch is 12 samples. We employ early stopping using a hold-out validation set (20% of training set), stopping training if the validation set’s localization error does not drop for 150 epochs. We employ data augmentations with a probability of 0.5, uniformly sampling from a continuous range $[\alpha, \omega]$: Random scaling [0.8, 1.2], translation [-0.07%, 0.07%], rotation $[-45^\circ, 45^\circ]$, shearing [-16, 16] and vertical flipping.

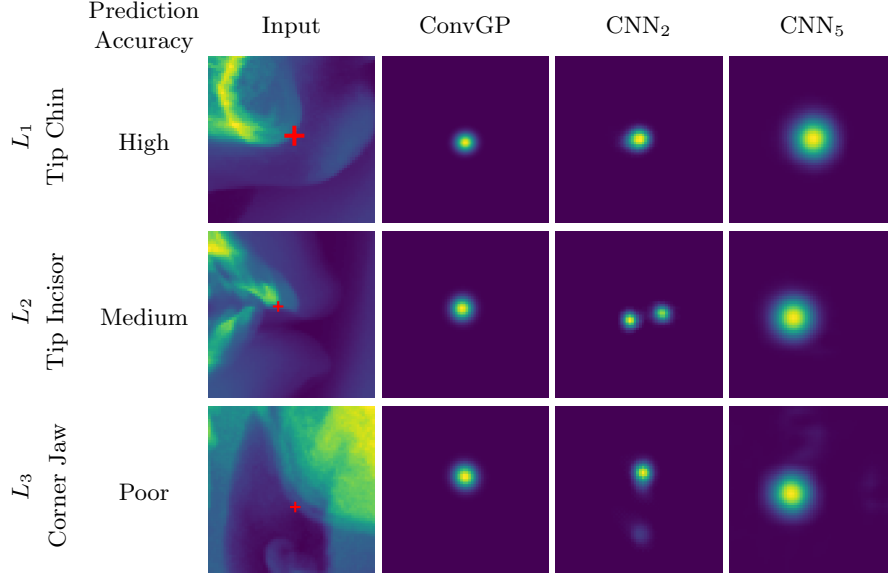


Fig. 2. Figure showing 3 landmarks of varying difficulty and predictions from our method with learned covariances (ConvGP), and 2 deep learning baselines with fixed covariances (CNN₅ and CNN₂).

Stage 2: We set $H' = W' = 64$ to select a 64×64 patches around the *Stage 1* predictions C , reaching a compromise between capturing enough context in the image patch and computational limitations. For the data augmentation, we select $D = 32$ for Eq. 5, enabling the model to train on a diverse set of images. We do not perform further data augmentation. At test time, we choose the model checkpoint with the lowest validation loss during training.

For the convolutional Gaussian process, we use a stride of 5 when extracting the 5×5 patches, since a stride of 1 is too demanding in terms of memory. We train for 6000 epochs with minibatches of size 6, using the Adam optimizer [11] for stochastic gradient descent with an initial learning rate of 0.01, reducing it after 3000 epochs to 0.001. To prevent the likelihood variance from growing large during optimization and dominating the posterior covariance, we fix it for 3000 epochs. For the inducing patch sampling, we set $\sigma = 1$ for the sampler in Eq. (4), heavily biasing the patch initializations to parts of the image near the *Stage 1* prediction. For the base kernel within the CGP, we use an automatic relevance determination (ARD) Matérn 1/2 kernel, initialized with a lengthscale of 1 for each input dimension, and a variance of 20.

For the CNN baseline, we use the same minibatch size, training length and learning rate schedule as the CGP. We experiment using $\sigma = 2$ and $\sigma = 5$ for Eq. (1) to compare the sensitivity of uncertainty estimates depending on the target heatmap size. The values respectively represent a high and low precision choice for the target heatmap, typically selected through a hyper-parameter search.

Table 1. Localization results from 3 landmarks of the Cepalmetric dataset [23] over a 4-fold CV. The Negative Log Predictive Density is reported (NLPD, lower is better), the mean point-to-point error (PE), in millimeters. Our non deep-learning (DL) method, convolutional Gaussian process (CGP), is compared to two DL baseline methods: CNN₂, CNN₅, which use a heatmap label $\sigma = 2$, $\sigma = 5$, respectively.

Metric		CGP	CNN ₂	CNN ₅
		(Learned Covariance)	(Fixed Covariance)	(Fixed Covariance)
All	NLPD(↓)	5.54 ± 2.24	6.83 ± 17.64	5.51 ± 1.87
	PE (↓)	1.81 ± 1.11	1.36 ± 1.84	1.15 ± 1.28
L ₁	NLPD (↓)	4.67 ± 1.33	4.30 ± 3.22	5.25 ± 0.23
	PE (↓)	1.26 ± 0.76	0.91 ± 0.75	0.87 ± 0.58
L ₂	NLPD (↓)	5.35 ± 1.73	5.50 ± 14.14	5.27 ± 0.85
	PE (↓)	1.72 ± 1.00	0.84 ± 1.58	0.68 ± 0.93
L ₃	NLPD(↓)	6.57 ± 3.55	10.67 ± 35.56	6.02 ± 4.52
	PE (↓)	2.46 ± 1.55	2.33 ± 3.19	1.90 ± 2.34

3.4 Results and Discussion

Table. 1 shows that our CGP is capable of reliably quantifying predictive uncertainty, achieving an NLPD of 5.54. The CNN with a target heatmap of $\sigma = 5$ (CNN₅) achieves a marginally lower NLPD score of 5.51, despite achieving much better localization error of 1.15mm compared to the CGP error of 1.84mm. Notably, an improved mean estimate (PE) will in turn improve the NLPD. Therefore, the fact that the CGP achieves an NLPD within approximately 0.03 of CNN₅ is encouraging.

Moreover, the performance of the CNN model, in terms of both the PE and NLPD, is highly dependent on the hyperparameter σ , resulting in significant variations between CNN₅ and CNN₂. This reflects the limitations of deep learning methods in measuring uncertainty, as the NLPD score is almost entirely dependent on the mean estimate. This is exemplified by the results of CNN₂ on L₃, where a poor PE of 2.33mm leads to extremely poor uncertainty estimations, resulting in an NLPD score of 10.67. In contrast, the CGP, with a higher PE error of 2.46mm, still manages to produce more reliable uncertainty estimates, achieving an NLPD score of 6.57. Unlike the CNNs, the covariance of the CGP is learned during training and mathematically grounded, making it a more trustworthy and stable measure of the model’s confidence.

Figure 2 highlights how the distribution of the CNN’s output heatmap is highly dependent on the value of σ from the target function in Eq. (1), and is uniform across all landmarks. Note, that CNN₂ even has multiple hallucinated covariances. In contrast, the covariances learned by the CGP are distinct for each landmark, but unfortunately show a relatively uniform pattern across predictions of the same landmark. This uniformity is due to the fact that the covariance function was dominated by the likelihood noise during training, resulting in more homogeneous uncertainty estimates than optimal.

4 Conclusion

We showed that convolutional Gaussian processes (CGPs) can be applied to the complex vision task of landmark localization, allowing us to quantify the uncertainty associated with predictions using a nonparametric approach. Empirically, the localization error obtained with CGPs are not yet competitive with those of a CNN. This is attributed to optimization challenges related to the GP likelihood noise, which dominates the output covariance matrices, resulting in relatively uniform uncertainty estimates. Despite this, the CGP obtains a similar NLPD to the CNN baseline, suggesting that if this optimisation issues can be addressed and the predictive error decreased, the uncertainty quantification provided by the CGP would be far superior to that of the CNN. Furthermore, the uncertainty estimates provided by GPs are more rigorous mathematically and more interpretable in practice than those of deep learning. A promising avenue for future work could address the uniformity of the uncertainty estimates by incorporating a heteroscedastic likelihood that produces different likelihood variance outputs for each image.

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